



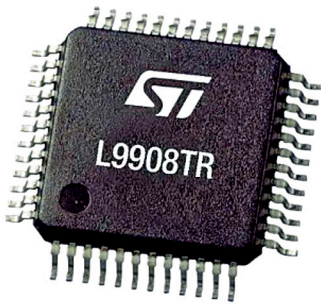
LTE-Cat.1bis standard. It provides a low-cost solution and simplifies design, thereby reducing development cost and time. The ELS62 module integrates security features to protect both the device and data, by ensuring robustness against cyberattacks, with strong device authentication and data encryption. It can be used in applications such as medical wearables, smart meters, and home alarm systems.

Thales

<https://www.thalesgroup.com/en>

3-phase GDU for 48V automotive applications

ST Microelectronics L9908 is a 3-phase gate driver unit (GDU) developed for 48V automotive applications. The GDU



is capable of controlling 6 N-channel FETs for brushless motors. It has flexible I/O channels and three current sensors. The module is capable of operating in 12V, 24V, and 48V systems. It features a 32-bit SPI interface to offer diagnostic and protection features, such as real-time phase voltage monitoring along with open load detection and a few others.

ST Microelectronics

<https://www.st.com>

Ultra-sensitive capacitive barometric pressure sensor

The BMP581 is a barometric sensor from Bosh Sensortec that detects fluctuations equivalent to one-thousandth of a mosquito's weight. The energy-efficient sensor is a highly accurate altitude tracking IC that can provide precise location information for appli-

cations such as indoor localisation, floor detection, and navigation. The portable size and low power consump-



tion make it suitable for smart wearables, hearables, and IoT applications.

Bosh Sensortec

<https://www.bosch-sensortec.com>

New-generation 3D touchless user interfaces

TouchNetix has developed fully integrated aXiom touchscreen chips that offer new 3D sensing capabilities by detecting air gestures, allowing



touchless functions in a variety of automotive, industrial, and consumer environments. TouchNetix' single-chip aXiom technology enables proximity sensing 10-15cm above the surface of the screen and provides much higher signal-to-noise ratio (SNR) than the traditional touchscreen controllers available in the market.

TouchNetix

<https://www.touchnetix.com>

Energy harvesting and power management in same chip

Diatom NH16D3045 is a high-performance energy harvester with an integrated power management

solution for low-power applications. The IC can be used to charge a variety of energy storage elements, such as



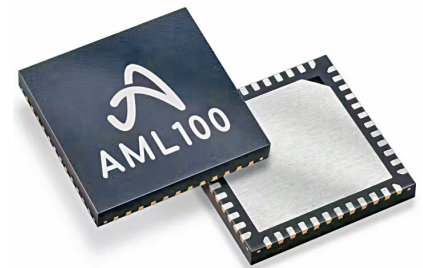
rechargeable batteries or supercapacitors, while it can harvest energy from light or from all types of solar PVs and from other energy sources, including kinetic, thermal, and radio frequencies. It is designed to work in the range of microwatts to a few milliwatts of power suitable for devices like smart wearables, electronic shelf labels, and wireless sensor networks.

Diatom

<https://www.nowi-energy.com>

World's first fully analogue machine learning chip

Aspinity's AML100 is the first product in Aspinity's AnalogMLfamily that can detect and classify sensor-driven events from raw, analogue sensor data. It allows developers to design significantly lower-power, always-on edge



processing devices. The IC consumes less than 20µA when in always-sensing, thus, enabling a ultra-low power always-on edge-processing solution for voice-first systems, acoustic event monitoring applications, predictive and preventative maintenance for industrial equipment, and biomedical monitoring.

Aspinity

<https://www.aspinity.com>